

RF and Microwave Engineering (EX752 / EX716)

Concise PYQ

July 22, 2026

Notes: TU IOE PYQ 2069–2081. Regular = **bold**, Back = normal; BEX = normal font, BEI (EX716) = **mono**. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Microwave frequency bands (IEEE) + applications; comm protocols (4)(2+4)(3+3)(2+6)(3+3+4)
- Conventional Vs RF/microwave circuits; lumped Vs distributed (4+2)(3+3)
- (+)(-) of microwaves Vs acoustic / conventional / lower-freq (5)(6)(2+4)
- Seismic band Vs RF/microwave circuit (8)

2. RF & M/W Transmission Lines

- Microwave TL Vs conventional; microstrip Vs strip-lines (4)(5)
- Admittance / immittance chart; sketch + duality parameters (2)(10)(4)
- Single-stub matching ($Z_L=80+j100$; $\Gamma_L=0.5\angle 51^\circ$) +Smith (2+8)(8)
- Double-stub matching ($3\lambda/8$) +Fig/+S-matrix; various loads (8)(10)(8+2)
 - $Z_L=300+j300$, $Z_L=80+j180@3\text{GHz}$, $Z_L=190+j110@10\text{GHz}$
 - $Z_L=60-j80/50\Omega$, $109+j120/75\Omega$ coax, $\Gamma=0.64\angle 58/100\Omega$
 - admittance $0.00813+j0.0065\Omega/50\Omega$; short-ckt $75+j75$, $0.4+j0.85$

3. RF & M/W Network Theory

- S-parameters (I); Vs h-parameters; return & insertion loss (4)(4+4)(2)
- S-parameters — two-port (derive); three-port network +Eg (2+6)(4+5)(4+4)
- S-matrix analysis: reciprocal / lossless / return loss / reflected power (3+1+1+1+2)
- Passive device from S-matrix (4-port); power splitter; E/H-Tee & Duplexer S-matrices (6)(8)

4. RF/Microwave Components & Devices

- Waveguide — rect TE/TM field equations; modes + parameters (7+3)(8+2)(10)
 - TE_{10} Vs TE_{20} cutoff; prove $\text{TM}_{01}/\text{TM}_{10}$ absent; dominant mode (2)(3)(5)
 - circular waveguide TE/TM field; air-filled rect (cutoff/velocities/ Z_0) +directional coupler (10)(6+4)
- Tee junctions — Magic Tee (S-params, +derive); E-plane Vs H-plane; hybrid tee (3+3)(4)(5)(8)
- Directional couplers; circulators; cavity resonators; microstrips (6)(5)(6)(5)

- Probe- Vs loop-coupling; Gunn diode; MASER (5)(4)

5. Microwave Generators

- Transit-time effect + two-cavity klystron; bunching (density modulation) (2+8)(2)
- Klystron oscillator (multi-cavity); velocity modulation + reflex klystron +Fig (2+7)(5)(10)
- Magnetron construction/operation; cross-field (45/60 phase) as power amp (2+8)(10)(6+2)(8)
- TWT construction & operation; conventional-tube limits at microwave; BWO; LNA (2+6)(2+8)(5)

6. RF Design Practices

- Filter — insertion loss method; Butterworth/Chebyshev prototype (5)(8)(4)
- LPF/HPF via microstrip (π -section, T-type, double-section shunt) +prototyping (3+5)(6+4)(10)(7+3)
- Amplifier — stability + max gain (bilateral & unilateral), +Smith, +flowchart (5+5)(10)(12)(18)
 - S-params: $0.894\angle -60.6^\circ$; $0.55\angle 150^\circ$; $0.656\angle 146.7^\circ$; $0.72\angle -116^\circ$ (GaAs); $0.64\angle -169^\circ$; $0.55\angle -160^\circ$ (MESFET)
 - flowchart (GaAsFET) + Γ_{in}/Γ_{out} trace; amplifier Vs oscillator (stability factor) (4)(5+5)
- Oscillator & mixer theory (5)(8)

7. Microwave Antennas & Propagation

- RF/MW radiation hazards + fields; safety practices & standards (SARPs) (8)(5)(6)(3+3)
- Effect of SAR; how radiation is hazardous; GSM BTS propagation fields + SAR (6)(5)(4+4)
- Antenna as DUT — measurement tool + working principle; HRP (8)(4)

8. RF/Microwave Measurements

- Power measurement — VSWR meter + low power / bolometry; VSWR > 10 (2+8)(4+6)(2+6)
- Calorimeter — static / circulating Vs flow; calorimeter wattmeter (5)(2+8)(6+2)
- Bolometer bridge (single-bridge limits, double-channel); network analyzer; two-port model (8)(4)(10)
- Microwave Vs low-frequency measurements; spectrum analyzer (3)(4)